

**FLASHNET**

**NetGate 1100 Documentation**

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## 1. Device Overview

The Netgate 1100 is a small-form-factor firewall/router appliance designed by **Netgate**.

### Hardware Features

- 3x 1GbE switch ports (WAN/LAN configurable)
- ARM-based CPU
- Console access via micro-USB
- Pre-installed pfSense Plus

### Default Settings

- LAN IP: 192.168.1.1
- Username: admin
- Password: pfsense

## 2. Device Information



***Fig 1.1 NetGate 1100 Appliance***

### 2.1 Physical Setup

- Step 1: Cabling

Connect WAN port to ISP modem.

Connect LAN port to PC or switch.

Power on device.

- Step 2: Access Web GUI

From a LAN-connected PC:

- <http://192.168.1.1>

Login with default credentials.

## 2.2 Physical Connectivity

To get started, connect the cables in this specific order:

1. **WAN Port:** Connect to your Internet Service Provider (ISP) modem.
2. **LAN Port:** Connect to your computer or a network switch.
3. **Console Port:** (As seen in your screenshot) Connect to your PC via Micro-USB to use PuTTY.
4. **Power:** Plug in the 12V adapter.
  - *Note: Wait 4–5 minutes for the system to boot fully.*

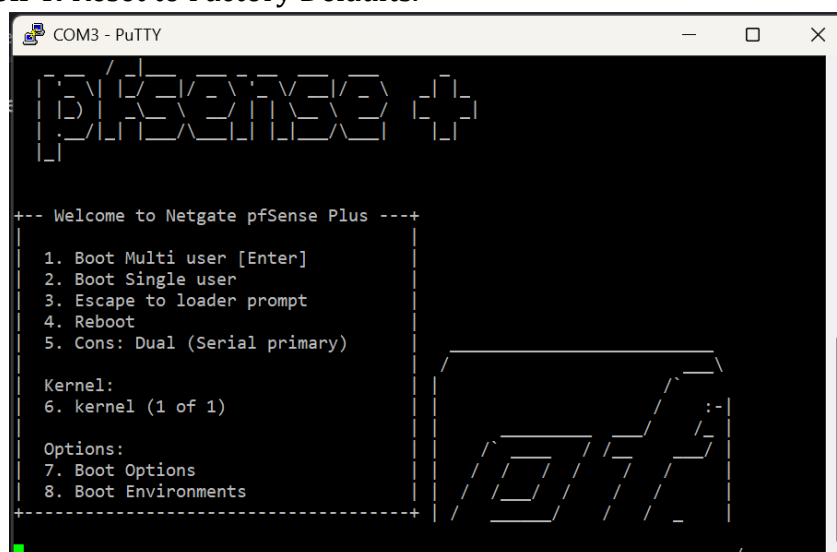
## 2.3 Initial Software Configuration

Once the boot loader menu (from your screenshot) finishes, you will see the **pfSense Console Menu**.

### 1. The Console Menu (Serial Access)

If you need to change settings before the web interface is accessible, use these common options:

- **Option 2:** Set Interface IP Address (Use this if you need to change the default 192.168.1.1).
- **Option 3:** Reset Admin Password (Useful if you're locked out).
- **Option 4:** Reset to Factory Defaults.

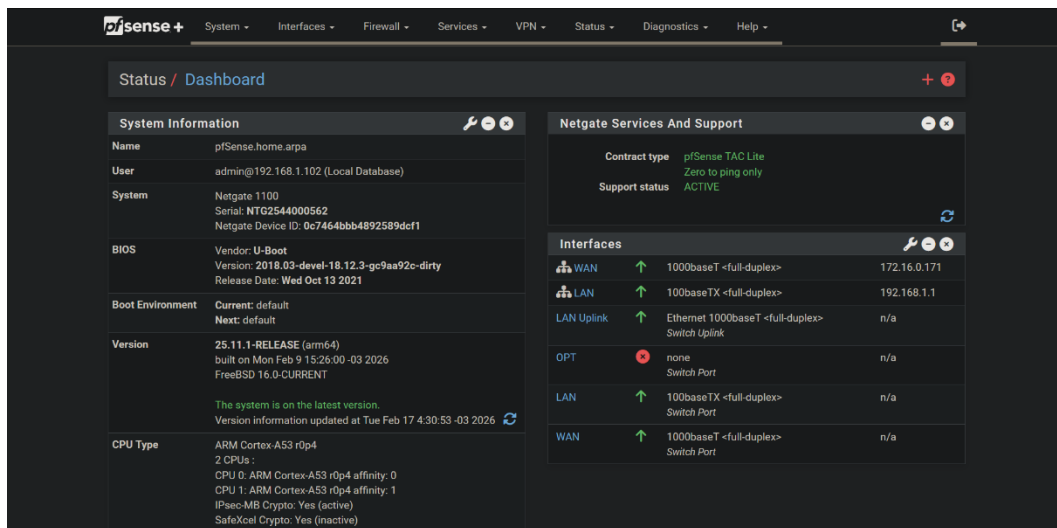


**Fig 1.2 PF Sense serial controller through PuTTY**

## 2. The WebGUI (The Main Interface)

Most users will manage the device here.

- **Default URL:** https://192.168.1.1
- **Default Username:** admin
- **Default Password:** pfsense



**Fig 1.3 PFsense through the webGUI**

## Section 3: Advanced Security Services (Packages)

The real power of the Netgate 1100 lies in its ability to run enterprise-grade security packages. In your demo, navigate to **System > Package Manager**.

- **pfBlockerNG:** This is the most popular add-on. It blocks ads, malicious domains, and entire countries (Geo-IP blocking) at the DNS level.
- **Snort or Suricata:** These are **IDS/IPS** (Intrusion Detection/Prevention Systems). They inspect packets for known "fingerprints" of hackers or malware.
- **Surveillance:** Explain that while the 1100 is an entry-level device, it can still handle light-duty traffic inspection for a home or small office.

## Section 4: Remote Access & Secure Connectivity

This is the "Work From Anywhere" section. Show how the Netgate 1100 creates a secure tunnel into the network.

### 1. WireGuard: The New Standard

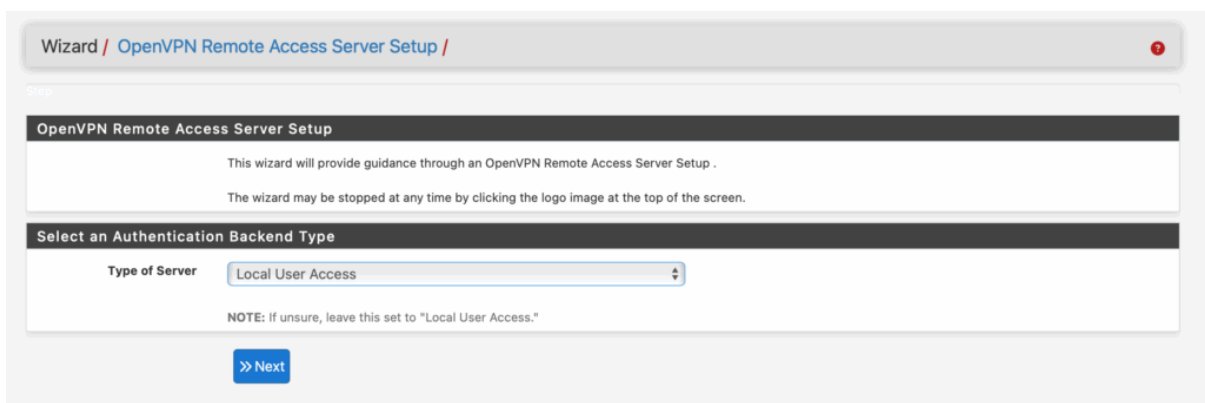
Navigate to **VPN > WireGuard**.

- **Speed:** Emphasize that WireGuard is incredibly "light" on the CPU. For the Netgate 1100's ARM processor, this is the most efficient way to get high-speed VPN.
- **Simplicity:** Show how a QR code can be generated to instantly set up a smartphone to connect back to the office securely.

### 2. OpenVPN with Client Export

Navigate to **VPN > OpenVPN > Client Export**.

1. This is a "Quality of Life" feature. Explain that pfSense makes a custom installer for Windows/Mac users.



**Fig 1.4 OpenVPN Setup through wizard**

### 3. Tailscale (Package)

Many users love Tailscale for its "Zero Config" nature.

- Mention that Tailscale can be installed as a package on the Netgate 1100, allowing it to act as a **Subnet Router** or **Exit Node**. This allows you to route all your phone's cellular traffic through your home's encrypted pfSense connection with zero port forwarding.

## Section 5: Firewall Rules & Logic

The firewall reads rules from the top of the list to the bottom. Once a packet matches a rule, it stops looking.

### 1. The Anatomy of a Rule

Navigate to **Firewall > Rules > LAN**. Show a rule and explain the three main actions:

- **Pass:** The green checkmark. "You are allowed through."
- **Block:** The red "X". The packet is dropped silently. (Best for WAN to hide your presence from hackers).
- **Reject:** The orange "X". The firewall tells the sender "No, go away." (Best for internal LAN to stop apps from "spinning" while waiting for a timeout).

### 2. Best Practice: The "Default Deny"

Explain that by default, pfSense blocks **everything** coming from the internet (WAN) but allows **everything** leaving your network (LAN).

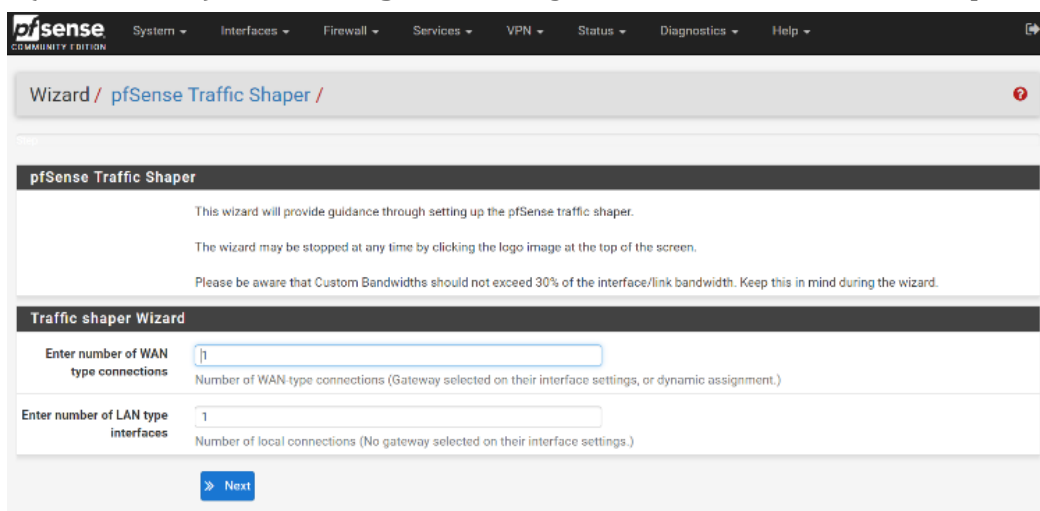
## Section 6: Traffic Shaping & Limiters

This is the "No More Lag" section. It's perfect for households where one person downloading a large game ruins the internet for everyone else.

### 1. The Traffic Shaper Wizard

Navigate to **Firewall > Traffic Shaper > Wizards**.

- Show the **Multi-WAN / Dedicated Link** wizard.
- It asks for your Upload/Download speeds and automatically prioritizes **VoIP (Phone calls)** and **Gaming** over "background" traffic like Windows Updates.



**Fig 1.5 Pfsense Traffic Shaper**

## 2. Bandwidth Limiters (The "Penalty Box")

Navigate to **Firewall > Traffic Shaper > Limiters**.

- Show how to create a "5Mbps Pipe."
- **Demo Scenario:** "If your roommate's BitTorrent is hogging the line, you can create a limiter and apply it to their specific IP. They will never be able to use more than 5Mbps, leaving the rest of the pipe open for your Zoom calls."

## Section 6: Firewall Schedules

Navigate to **Firewall > Schedules**. This is a favorite for parents or office managers.

- **How it works:** You define a time range (e.g., "Bedtime" from 10:00 PM to 6:00 AM).
- **Applying it:** Go back to your Firewall Rules, edit a rule for the Kids\_Devices alias, and select the "Bedtime" schedule.
- **The Result:** At 10:01 PM, the internet simply "turns off" for those specific devices, but stays on for yours.

**Fig 1.5 Assigning the configuration file for default browser**

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